

Problem 1. A discrete valuation on a field k is a map $v : k \rightarrow \mathbb{Z} \cup \{\infty\}$ such that $v(xy) = v(x) + v(y)$,

$$v(x + y) \geq \min\{v(x), v(y)\},$$

and

$$v(x) = \infty \iff x = 0.$$

Prove that the set

$$R := \{a \in k \setminus \{0\} \mid v(a) \geq 0\} \cup \{0\}$$

is a subring of k and is in fact a Euclidean domain.

(Rings of this form are called discrete valuation rings or DVRs).

Proof. For any $a, b \in R$,

$$[1] : v(1 \cdot a) = v(1) + v(a) = v(a) \implies v(1) = 0 \implies 1 \in R.$$

$$[+] : v(a + b) \geq \min\{v(a), v(b)\} \geq 0 \text{ since } a, b \in R \implies v(a), v(b) \geq 0 \implies a + b \in R.$$

$$[\cdot] : v(ab) = v(a) + v(b) \geq 0 \text{ since } a, b \in R \implies ab \in R.$$

$$[-] : v((-1)(-1)) = v(1) = v(-1) + v(-1) = 0 \text{ so } v(-1) = 0 \implies -1 \in R \text{ and then } (-1) \cdot a = -a \in R \text{ by } [\cdot].$$

So $R \leq K$ is a subring.

Now consider any $a, b \in R$, $b \neq 0$. Since K is a field, $ba^{-1} \in K$, and $v(ba^{-1}) = v(b) + v(a^{-1}) = v(b) - v(a)$. Therefore, if $v(a) \geq v(b)$, then $v(ba^{-1}) \leq 0 \implies ba^{-1} \notin R$. If $v(a) < v(b)$, then $v(ba^{-1}) > 0 \implies ba^{-1} \in R$. Therefore, if $v(a) \geq v(b)$, then $v(ab^{-1}) = v(a) - v(b) \geq 0 \implies ab^{-1} = \frac{a}{b} \in R$.

Also notice that $v(R) = \mathbb{N}$. We will use this to prove R is a Euclidean domain.

For any $a, b \in R \setminus \{0\}$, either

(i) $v(a) \geq v(b)$, or

(ii) $v(a) < v(b)$.

(i) If $v(a) \geq v(b)$, let $q = \frac{a}{b} \in R$, $r = 0$. We have $a = qb + r = \left(\frac{a}{b}\right)b + 0 = a$ with $r = 0$.

(ii) If $v(a) < v(b)$, then let $q = 0 \in R$ and $r = a \in R$. We have $a = qb + r = (0)b + a$ and $v(r) = v(a) < v(b)$.

So for all $a, b \in R \setminus \{0\}$, there exist $q, r \in R$ such that $a = qb + r$ where $r = 0$ or $v(r) < v(b)$ and $v(R) = \mathbb{N}$. Thus R is a Euclidean domain. $\square$

Problem 2. Let I, J be ideals in a commutative ring R , and define the maps

$$\varphi : R \rightarrow R/I \times R/J$$

by

$$r \mapsto ([r]_I, [r]_J)$$

and

$$\psi : R/I \times R/J \rightarrow R/(I+J)$$

by

$$([a]_I, [b]_J) \mapsto [a-b]_{I+J}.$$

Thus,

$$R \xrightarrow{\varphi} R/I \times R/J \xrightarrow{\psi} R/(I+J).$$

(a) Prove that $([a]_I, [b]_J) \in \text{im } \varphi$ if and only if there exists some $x \in R$ such that

$$[x]_I = [a]_I \quad \text{and} \quad [x]_J = [b]_J.$$

(b) Prove that $([a]_I, [b]_J) \in \ker \psi$ if and only if

$$[a]_{I+J} = [b]_{I+J}.$$

Proof. (a) Suppose $([a]_I, [b]_J) \in \varphi(R)$. Then there exists $x \in R$ such that $\varphi(x) = ([a]_I, [b]_J)$, so $([x]_I, [x]_J) = ([a]_I, [b]_J)$. So then $[x]_I = [a]_I$ and $[x]_J = [b]_J$.

Conversely, suppose there exists $x \in R$ such that $[x]_I = [a]_I$ and $[x]_J = [b]_J$. Then $\varphi(x) = ([x]_I, [x]_J) = ([a]_I, [b]_J)$, so $([a]_I, [b]_J) \in \varphi(R)$.

(b) Suppose $([a]_I, [b]_J) \in \ker \psi$. Then $\psi([a]_I, [b]_J) = [0]_{I+J}$, so $[a-b]_{I+J} = [0]_{I+J}$. Hence $a-b \in I+J$, which is equivalent to $[a]_{I+J} = [b]_{I+J}$.

Conversely, suppose $[a]_{I+J} = [b]_{I+J}$. Then $[a-b]_{I+J} = [0]_{I+J}$, so $\psi([a]_I, [b]_J) = [0]_{I+J}$. Therefore $([a]_I, [b]_J) \in \ker \psi$.

□

Problem 3. Using the same notation as Problem 2:

(a) Prove that

$$\text{im } \varphi = \ker \psi.$$

In this case, we say the sequence is exact.

Note that when $R = \mathbb{Z}$, this becomes the statement:

$$x \equiv a \pmod{n} \quad \text{and} \quad x \equiv b \pmod{m}$$

has a solution if and only if

$$a \equiv b \pmod{\text{lcm}(m, n)}.$$

(b) Find integers a, b, m, n such that

$$\gcd(m, n) \neq 1$$

but the system

$$x \equiv a \pmod{n}, \quad x \equiv b \pmod{m}$$

still has a solution.

Proof. (a) Suppose $([a]_I, [b]_J) \in \varphi(R)$. By the previous problem, there exists $x \in R$ such that $[x]_I = [a]_I$ and $[x]_J = [b]_J$. Hence $a - x \in I$ and $b - x \in J$. Therefore $a - b = (a - x) + (x - b) \in I + J$, so $[a]_{I+J} = [b]_{I+J}$. By the previous problem again, $([a]_I, [b]_J) \in \ker \psi$.

Conversely, suppose $([a]_I, [b]_J) \in \ker \psi$. Then $[a]_{I+J} = [b]_{I+J}$, so $a - b = i + j$ for some $i \in I$ and $j \in J$.

Let $x = a - i = b + j$. Then $x - a = -i \in I$ and $x - b = j \in J$, so $[x]_I = [a]_I$ and $[x]_J = [b]_J$. Hence $([a]_I, [b]_J) \in \varphi(R)$.

Therefore $\varphi(R) = \ker \psi$.

(b) Let $a = b = 0$ and $m = 2, n = 4$. (a) $\implies \begin{cases} x \equiv a \pmod{m} \\ x \equiv a \pmod{n} \end{cases} \iff a \equiv b \pmod{\text{lcm}(m, n)} = 2$.

This system has the solution $x = 8$ and $\gcd(2, 4) = 2 \neq 1$.

□

Problem 4. Review the notes on Euclidean domains, PIDs, and UFDs (Sections 1.13–1.15).

- (a) If R is a Euclidean domain and $I \subseteq R$, then we proved I is principal, say

$$I = (a)$$

for some $a \in R$. Describe a .

- (b) Outline the proof that every PID is a UFD. You do not need to reproduce the full proof, but list every major step.

Proof. (a) Let R be a Euclidean domain with Euclidean valuation $\delta : R \setminus \{0\} \rightarrow \mathbb{N}$, and let $I \subseteq R$ be a nonzero ideal.

Since $\delta(I \setminus \{0\}) \subseteq \mathbb{N}$, by the **Well-Ordering Principle** there exists $a \in I \setminus \{0\}$ such that $\delta(a)$ is minimal.

We prove that $I = \langle a \rangle$. Obviously, $\langle a \rangle \subseteq I$ since $a \in I$.

Now let $x \in I$. Since R is Euclidean, there exist $q, r \in R$ such that $x = qa + r$, where either $r = 0$ or $\delta(r) < \delta(a)$.

Since $x, a \in I$ and I is an ideal, $r = x - qa \in I$.

By minimality of $\delta(a)$, we can't have that $\delta(r) < \delta(a)$. So $r = 0$.

Therefore $x = qa \in \langle a \rangle$, so $I \subseteq \langle a \rangle$.

So each $I = \langle a \rangle$ where a minimizes the Euclidean evaluation δ among non-zero elements of I .

- (b) Let R be a PID.

We first prove existence of irreducible factorizations. Suppose there exists a nonzero non-unit which does not factor into irreducibles. Choose such an element x_1 .

Since x_1 is not irreducible, we may write $x_1 = x_2 y_2$, where neither x_2 nor y_2 is a unit. In particular, $x_1 \in \langle x_2 \rangle$, so $\langle x_1 \rangle \subseteq \langle x_2 \rangle$.

If equality held, then $x_2 \in \langle x_1 \rangle$, so $x_2 = r x_1$ for some $r \in R$. Hence

$$x_1 = x_2 y_2 = (r x_1) y_2 = x_1 (r y_2).$$

Since R is an integral domain and $x_1 \neq 0$, we get $1 = r y_2$, so y_2 is a unit, a contradiction.

Thus $\langle x_1 \rangle \subsetneq \langle x_2 \rangle$.

Continuing inductively gives a strictly increasing chain

$$\langle x_1 \rangle \subsetneq \langle x_2 \rangle \subsetneq \langle x_3 \rangle \subsetneq \cdots$$

But every PID is Noetherian, so such a chain cannot exist and every nonzero non-unit factors into irreducibles.

Now let p be irreducible and suppose $p \mid ab$. Consider the ideal $\langle p, a \rangle$. Since R is a PID, $\langle p, a \rangle = \langle d \rangle$ for some $d \in R$.

Since $d \mid p$ and p is irreducible, either d is a unit or d is associate to p .

If d is associate to p , then $\langle p, a \rangle = \langle p \rangle$, so $a \in \langle p \rangle$, i.e. $p \mid a$.

If d is a unit, then $\langle p, a \rangle = R$, so there exist $x, y \in R$ such that $1 = xp + ya$. Multiplying by b gives $b = xpb + yab$.

Since $p \mid ab$ and clearly $p \mid xpb$, it follows that $p \mid b$.

Therefore every irreducible is prime.

Finally, suppose $p_1 \cdots p_n = q_1 \cdots q_m$ with all p_i, q_j irreducible. Since p_1 is prime, $p_1 \mid q_1 \cdots q_m$, so $p_1 \mid q_j$ for some j . Since q_j is irreducible, p_1 is associate to q_j . After reordering and cancelling associates, continue inductively and you can find some associate q_j for each p_i .

So we see factorization is unique up to associates and reordering, and R is a UFD. □

Problem 5. Let R be an integral domain, and let p be a prime element of R .

- (a) Prove that p is not a unit.
- (b) Prove that if q is an associate of p , then q is also prime.

Proof. (a) Suppose p is a unit. Then $\langle p \rangle = \langle 1 \rangle = R$, which is not a proper ideal. Since a prime ideal must be proper, this is a contradiction. Therefore p is not a unit.

(b) Suppose q is associate to p . Then $q = up$ for some unit $u \in R^\times$.

Now suppose $q \mid ab$. Then $ab = qk = upk$ for some $k \in R$, so $p \mid ab$.

Since p is prime, either $p \mid a$ or $p \mid b$.

If $p \mid a$, then $a = pr$ for some $r \in R$. Since $p = u^{-1}q$, we have

$$a = pr = (u^{-1}q)r = q(u^{-1}r),$$

so $q \mid a$.

Similarly, if $p \mid b$, then $q \mid b$.

Therefore q is prime. □

Problem 6. Let R be a commutative ring and let $S \subseteq R$ be a multiplicative subset of R .

(a) Prove that if I is an ideal of R such that

$$I \cap S = \emptyset,$$

then

$$I^e := S^{-1}I = \left\{ \frac{a}{b} \mid a \in I, b \in S \right\}$$

is a proper ideal of $S^{-1}R$.

(b) Let

$$\varphi : R \rightarrow S^{-1}R$$

be the natural ring homomorphism defined by

$$r \mapsto \frac{r}{1}.$$

Prove that if J is a proper ideal of $S^{-1}R$, then

$$J^c := \varphi^{-1}(J)$$

is an ideal of R such that

$$J^c \cap S = \emptyset.$$

Proof. (a) $\forall \frac{i_1}{s_1}, \frac{i_2}{s_2} \in S^{-1}I$, we have that $i_1, i_2 \in I$ and $s_1, s_2 \in S$. So then $s_2 i_1 - s_1 i_2 \in I$ and

$$\frac{i_1}{s_1} + \frac{i_2}{s_2} = \frac{s_2 i_1 + s_1 i_2}{s_1 s_2} \in S^{-1}I \leq_+ S^{-1}R.$$

$\forall \frac{r}{s} \in S^{-1}R$ we have

$$\frac{r}{s} \cdot \frac{i_1}{s_1} = \frac{r i_1}{s s_1} \text{ and since } I \text{ is an ideal, } r i_1 \in I, \text{ so } \frac{r i_1}{s s_1} \in S^{-1}I.$$

Therefore $S^{-1}I$ is an ideal of $S^{-1}R$.

(b) Since $\varphi : R \rightarrow S^{-1}R$ is a ring homomorphism and J is a proper ideal $J^c = \varphi^{-1}(J)$ is an ideal of R . Now suppose $s \in J^c \cap S$. Then $\varphi(s) = \frac{s}{1} \in J$.

But

$$\frac{s}{1} \cdot \frac{1}{s} = \frac{1}{1},$$

so $\frac{s}{1}$ is a unit in $S^{-1}R$. Since $\frac{s}{1} \in J$ and $\frac{1}{s} \in S^{-1}R$ we get $\frac{1}{1} = 1_{S^{-1}R} \in J$ and so $J = S^{-1}R$, contradicting that J is proper.

Thus $J^c \cap S = \emptyset$.

□

Problem 7. Using the same notation as Problem 6, let J be a proper ideal of $S^{-1}R$.

(a) Prove that

$$(J^c)^e = J$$

while

$$(I^e)^c = \{a \in R \mid (\exists s \in S) \text{ such that } sa \in I\}.$$

(b) Find an example showing that $(I^e)^c$ need not equal I , even if $I \cap S = \emptyset$.

(Hint: Let

$$S = \{1, x, x^2, \dots\}$$

in $R = \mathbb{C}[x, y]$, and consider $I = \langle xy \rangle$.)

Proof. For Problem 7:

(a) First we prove $(J^c)^e = J$. Since $J^c = \varphi^{-1}(J)$, every $a \in J^c$ satisfies $a/1 \in J$. So every element of $(J^c)^e$ is of the form $(r/s)(a/1)$ with $a/1 \in J$ and $(J^c)^e \subseteq J$.

Conversely, let $a/s \in J$. Since J is an ideal and $s/1 \in S^{-1}R$, we have $(s/1)(a/s) = a/1 \in J$. So $a \in J^c$, and $a/s = (1/s)(a/1) \in (J^c)^e$. Therefore $J \subseteq (J^c)^e$, and $(J^c)^e = J$.

Now we prove $(I^e)^c = \{a \in R \mid (\exists s \in S) \text{ such that } sa \in I\}$.

Let $a \in (I^e)^c$. Then $a/1 \in I^e = S^{-1}I$. Hence $a/1 = i/t$ for some $i \in I$ and $t \in S$. By equality in the localization, there exists $u \in S$ such that $u(ta - i) = 0$. Thus $uta = ui \in I$. Since $ut \in S$, this shows there exists $s \in S$ such that $sa \in I$.

Conversely, suppose there exists $s \in S$ such that $sa \in I$. Then $a/1 = sa/s \in S^{-1}I = I^e$ and $a \in (I^e)^c$. Therefore $(I^e)^c = \{a \in R \mid (\exists s \in S) \text{ such that } sa \in I\}$.

(b) Let $R = \mathbb{C}[x, y]$, let $S = \{1, x, x^2, \dots\}$, and let $I = \langle xy \rangle$. Then $I \cap S = \emptyset$, since no power of x is divisible by xy .

However, $y \in (I^e)^c$, because $xy \in I$ and $x \in S$. But $y \notin I = \langle xy \rangle$. Therefore, $(I^e)^c \neq I$. $\square$

Problem 8. Let R be a UFD, and let S be a multiplicative subset of R .

- (a) Prove that if q is irreducible in R , then $q/1$ is either irreducible or a unit in $S^{-1}R$.
- (b) Prove that if a/s is irreducible in $S^{-1}R$, then a/s is associate to $q/1$ for some irreducible element q of R .
- (c) Prove that $S^{-1}R$ is also a UFD.

Proof. (a) Let q be irreducible in R . Since R is a UFD, q is prime.

If $q/1$ is a unit in $S^{-1}R$, then we are done. So suppose $q/1$ is not a unit. We prove $q/1$ is irreducible.

Suppose $q/1 = (a/s)(b/t)$ in $S^{-1}R$. Then, after clearing denominators, there exists $u \in S$ such that $uqst = uab$. Since R is a domain, $qst = ab$. Since q is prime, $q \mid a$ or $q \mid b$.

If $q \mid a$, we have $a = qa'$ for some $a' \in R$. Then $q/1 = (qa'/s)(b/t)$, and then cancelling $q/1$ gives $1 = (a'/s)(b/t)$. So b/t is a unit. Similarly, if $q \mid b$, then a/s is a unit. Therefore $q/1$ is irreducible in $S^{-1}R$.

(b) Suppose a/s is irreducible in $S^{-1}R$. Factor a in the UFD R as $a = uq_1 \cdots q_n$, where $u \in R^\times$ and each q_i is irreducible in R .

Then in $S^{-1}R$, $a/s = (u/1)(q_1/1) \cdots (q_n/1)(1/s)$. Here $u/1$ and $1/s$ are units. Since a/s is irreducible, exactly one of the factors $q_i/1$ is nonunit, and all the others are units. Therefore a/s is associate to $q_i/1$ for some irreducible $q_i \in R$.

(c) Every nonzero nonunit $a/s \in S^{-1}R$ factors into irreducibles by factoring a in R and discarding the factors which become units in $S^{-1}R$.

By part (b), every irreducible in $S^{-1}R$ is associate to $q/1$ for some irreducible $q \in R$.

Also, if $q/1$ is irreducible in $S^{-1}R$, then it is prime. Suppose $(q/1) \mid (a/s)(b/t)$. Clearing denominators gives $wuab = wqcst$ for some $w, u \in S$. Since q is prime in R , q divides one of w, u, a, b, s, t . But if q divided one of $w, u, s, t \in S$, then $q/1$ would be a unit in $S^{-1}R$, contradiction. So $q \mid a$ or $q \mid b$, and then $q/1$ divides a/s or b/t .

Thus irreducibles in $S^{-1}R$ are prime. The usual prime-divisibility argument gives uniqueness of factorizations up to associates and order. Therefore $S^{-1}R$ is a UFD.

□